

Editor Reference Document

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GAM250F20
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- Meetings and Impediments / Roadmap and Milestone / Deliveries

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- Rendering / Graphics Pipeline

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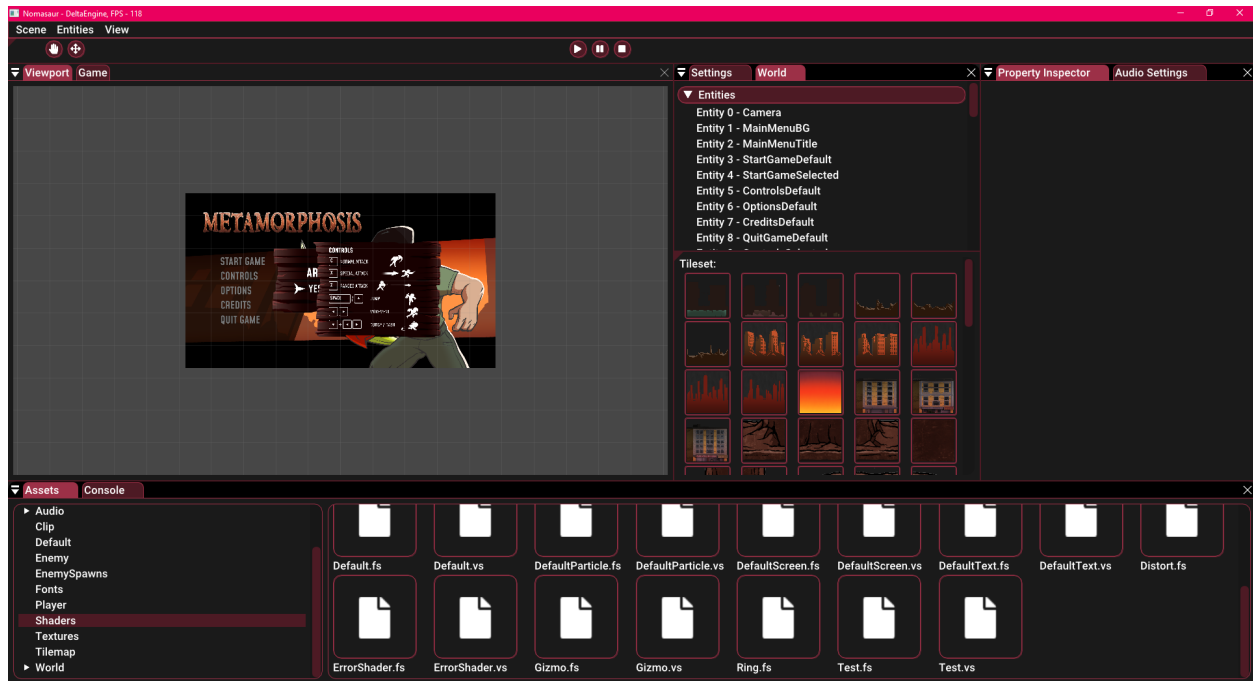
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- Physics / Collision / AI / Gameplay

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Editor's Overview



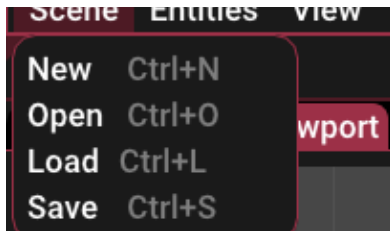
The editor tool is designed with docking capabilities, where each panel is specialized to handle a set of requirements needed to effectively create the game in the custom engine.

The following are the panels currently available in the editor:

- Animator Panel
- Assets Panel
- Audio Panel
- Bezier Panel
- Buttons Panel
- Game Panel
- Property Inspector Panel
- Settings Panel
- Sprite Editor Panel
- Tilemap Panel
- Viewport Panel
- World Panel

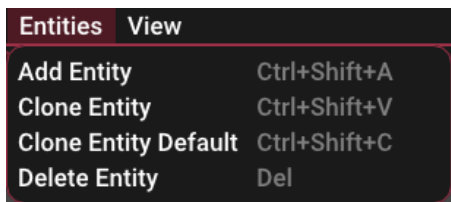
Editor's Menu Bar

Located at the top left corner of the editor window.



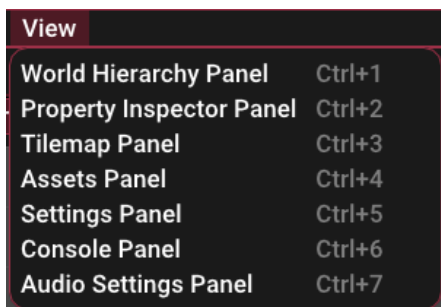
Scene

- New (Ctrl+N)
 - Create a new world
- Open (Ctrl+O)
 - Open a new world file
- Load (Ctrl+L)
 - Stop loading the current world file and load another world file
- Save (Ctrl+S)
 - Save the current world file



Entities

- Add Entity (Ctrl+Shift+A)
 - Create a new entity in the world
- Clone Entity (Ctrl+Shift+V)
 - Create an exact copy of the entity in the world
- Clone Entity Default (Ctrl+Shift+C)
 - Create a entity with the same archetype as the current selected entity



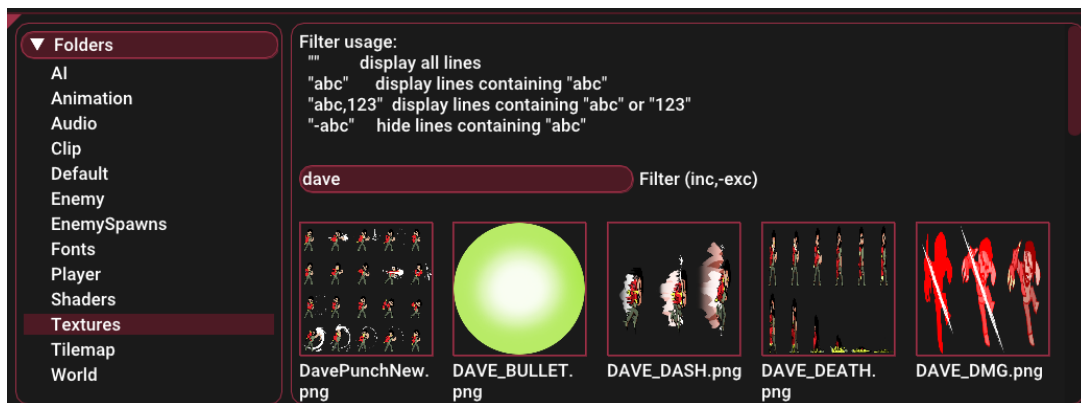
View

- World Hierarchy Panel (Ctrl+1)
 - Enable World Panel
- Property Inspector Panel (Ctrl+2)
 - Enable Property Inspector Panel
- Tilemap Panel (Ctrl+3)
 - Enable Tilemap Panel
- Assets Panel (Ctrl+4)
 - Enable Assets Panel
- Settings Panel (Ctrl+5)
 - Enable Settings Panel
- Console Panel (Ctrl+6)
 - Enable Console Panel
- Audio Settings Panel (Ctrl+7)
 - Audio Settings Panel

[illegible]

- Parameters
 - Add will create a new parameter, with the functionality to name/rename the parameter.
 - Selecting a parameter will expand its field, usually the fields are left empty and modified in the logic system.
 - Clicking on Delete will delete the parameter.
- Visualizer
 - Holding middle mouse button will enable you to drag and move around the visualizer
 - Click on a Clip will open it up for modification in the inspector
 - You can right-click on a clip
 - Make Transition
 - After clicking the “Make transition” option, you can select the end point of this transition, this will create a new transition connecting the current clip to another clip through the transition.
 - The new transition can be modified in the inspector panel
 - Delete
 - Delete the clip
 - Set as Entry
 - Set the clip chosen as the entry point
- Inspector
 - Change the selected clip
 - Click on transition to modify the transiting condition, add or set. Delete condition has not been implemented yet

Assets Panel



This panel is used to browse all the assets mounted by the engine's system directories. These assets will be loaded by the engine, hence available to the game when run.

The left side is the folders panel, the right is the file panel in the selected directory. Folder will show up first.

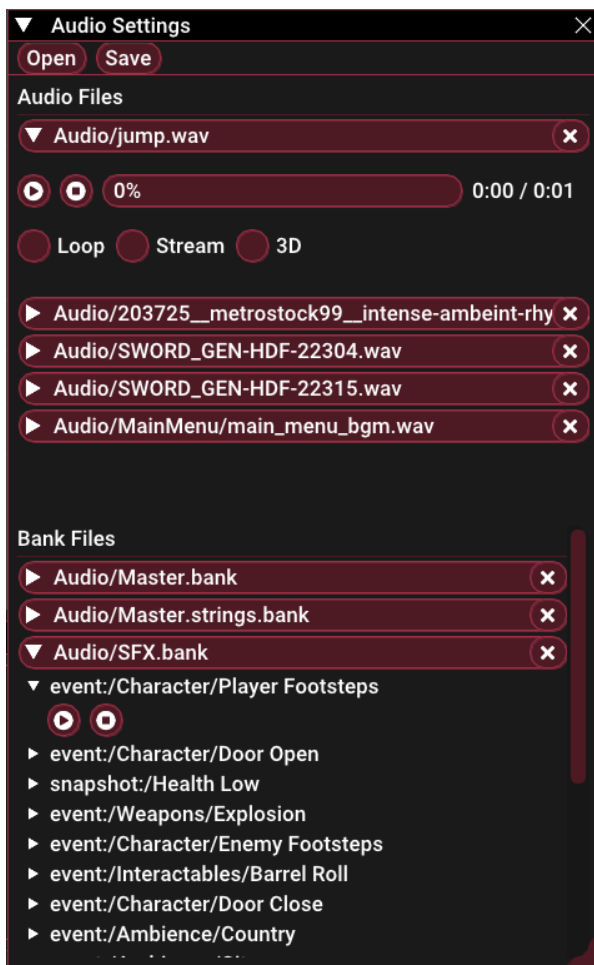
- Drag & Drop from file explorer
 - Assets file from the file explorer can be dragged and dropped into the asset panel. This will copy those files into the current selected directory
- Drag & Drop from Asset Panel to:
 - Viewport Panel
 - Dragging a texture will create an entity using that texture in the world.
 - Audio Settings Panel
 - Dragging a sound / bank files will add it into the configuration
 - Property Inspector Panel
 - Animator -> Controller Key field
 - Dragging an .anim file will update the animation key
 - Image -> Sprite field
 - Dragging a texture will update the texture
 - Animator Panel
 - Dragging a clip will allow for transition to that clip for animation
- Filter
 - Typing a name will filter the files that are displayed
- Double-Click files:
 - .anim: Opens up the animator editor to edit that .anim file
 - Textures: Opens up the sprite editor to edit that texture file

Audio Settings Panel

The audio panel can be opened from the menu bar -> View -> Audio Settings Panel, or shortcut (Ctrl+7).

As the engine supports both FMOD Core & FMOD Studio, the audio settings are designed to handle both types of files.

Only audio files & bank files that are setup in the audio settings will be loaded by the engine and used in the game.



Open

- Opens a file dialog where user can choose another audio settings file

Save

- Will save the current settings in a default file called "Settings.cfg" which will be used by the engine when runned

Audio Files

- Drag & Drop
 - Drag audio file from asset panel into this area to add it into the configuration
- Play / Pause
- Loop / Stream / 3D
 - Audio file mode

Bank Files

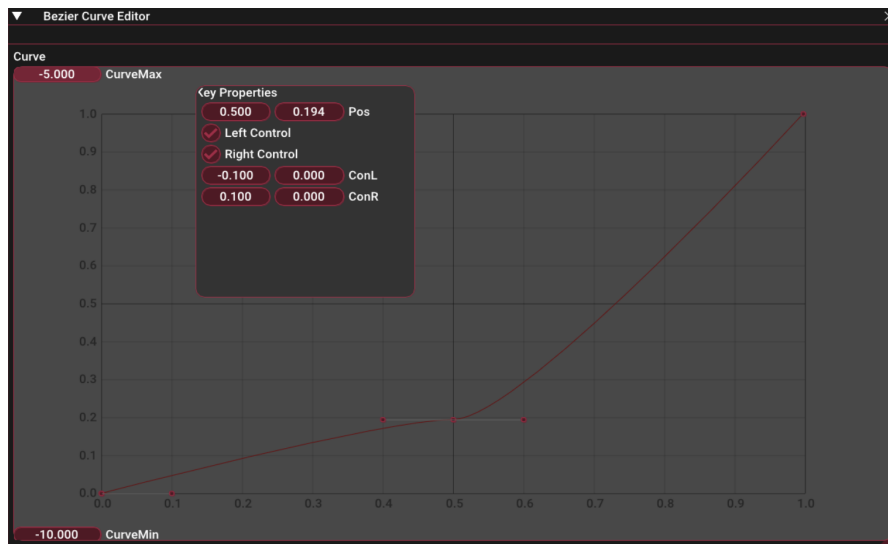
- Play / Pause
 - Only events can be played

Audio and bank files must be added into the Engine through the audio panel to be used in the Editor or the game. If not, the selection list will not be available in the Property Inspector Panel.

No modification capability is allowed for bank files as these specification are set-up in FMOD Studio DAW

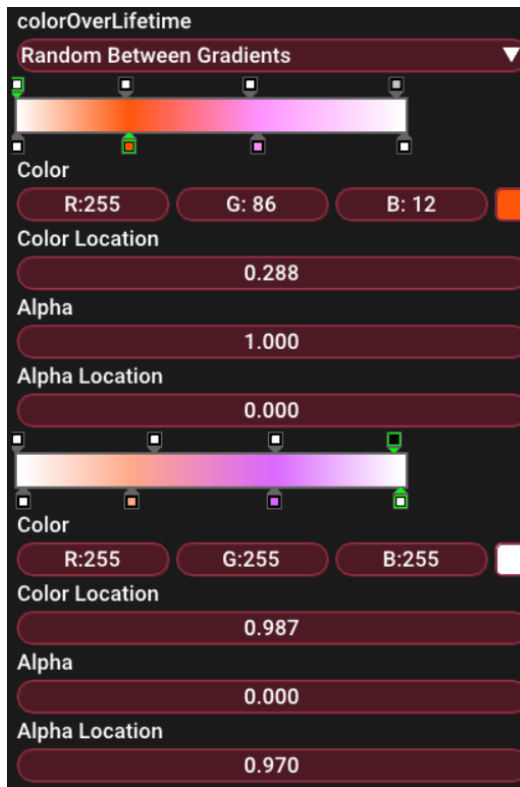
Bezier Panel

The particle emitter editor in the property inspector is used to edit the particle emitter component. This component has many data members, and some of their types are complex and have special panels for them.



Bezier Curve Editor

- Supports 4 keys in the curve that can be dragged around the graph to move them.
- Their positions are clamped to be between 0 to 1, the time value for interpolation between the min and max.
- Key properties can be edited to



Gradient Editor

- Supports 4 keys for alpha and color values.
- Their locations are clamped to be between 0 to 1. The values are clamped to the standard 0 to 1 as well, with color being displayed in hexadecimal.
- Multiple modes are available: constant color, random between colors, constant gradient, random between gradients.

Buttons Panel



The button panel is located at the top of the editor just right below the menu bar.

This panel contains the following button:



Hand Tool: Must be selected to allow dragging to move the camera in the viewport panel



Entity Selection Tool: Must be selected to allow selection of entity in the viewport panel



Play: Run the game simulation, currently it is required to save before pressing play as this will set the world to run. In the future, this button will create a duplicate world and run it in the game panel.



Pause: Pause the game simulation.



Stop: Stop the game simulation by destroying the duplicated world.



Toggle mouse cursor: Toggle mouse cursor on or off.

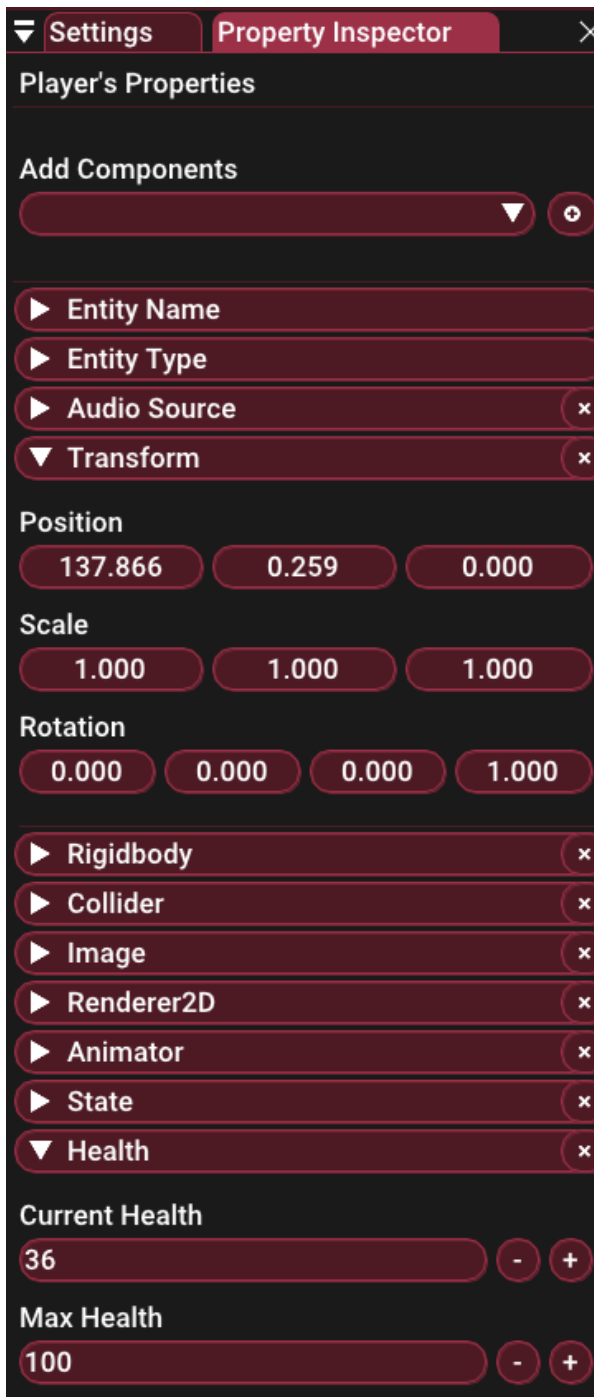
Game Panel



The game panel is used to display the game's camera, so that designers can play the game while editing to test things out.

Property Inspector Panel

The property inspector panel is used to view the components that are attached to the currently selected entity, allowing modification of the component data.



Add Components

- Click on the dropdown list to view the list of available components that can be added to the entity.
- After selection, the “+” button will add the selected component to the entity

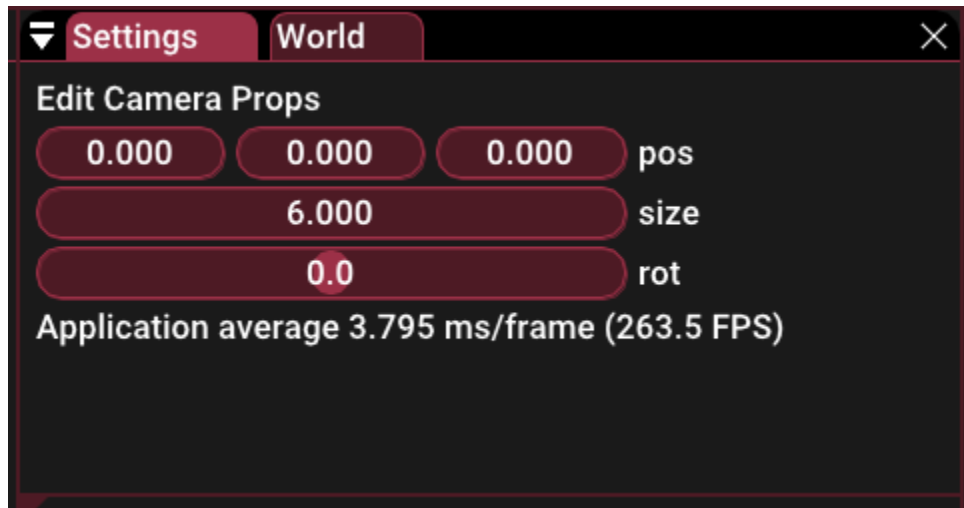
Components Modification

- Clicking on components will expand the properties that can be configured for that component.

Required Component Pairing

- For physics to run
 - Rigidbody
 - Collider
- For Animation to run
 - Animator
 - State
 - Renderer2D
 - Image
- For Graphics
 - Renderer2D
 - Image
- For Text
 - Renderer2D
 - Text
- For UI
 - RenderOverlay
 - Image
- For GUI
 - RenderOverlay
 - Image
- For AI
 - AI
 - Serpentipede/Fiddler/Lancer

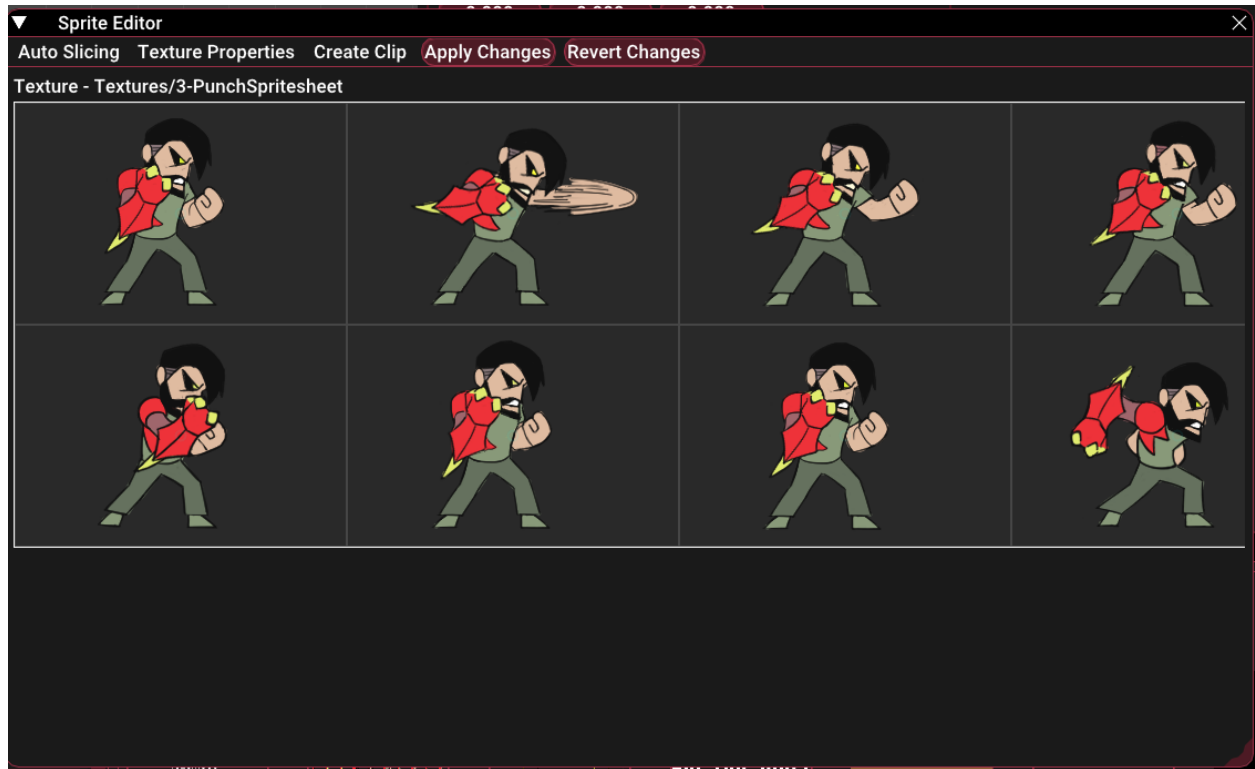
Settings Panel



This panel is used for modifying the Editor's Viewport Position.

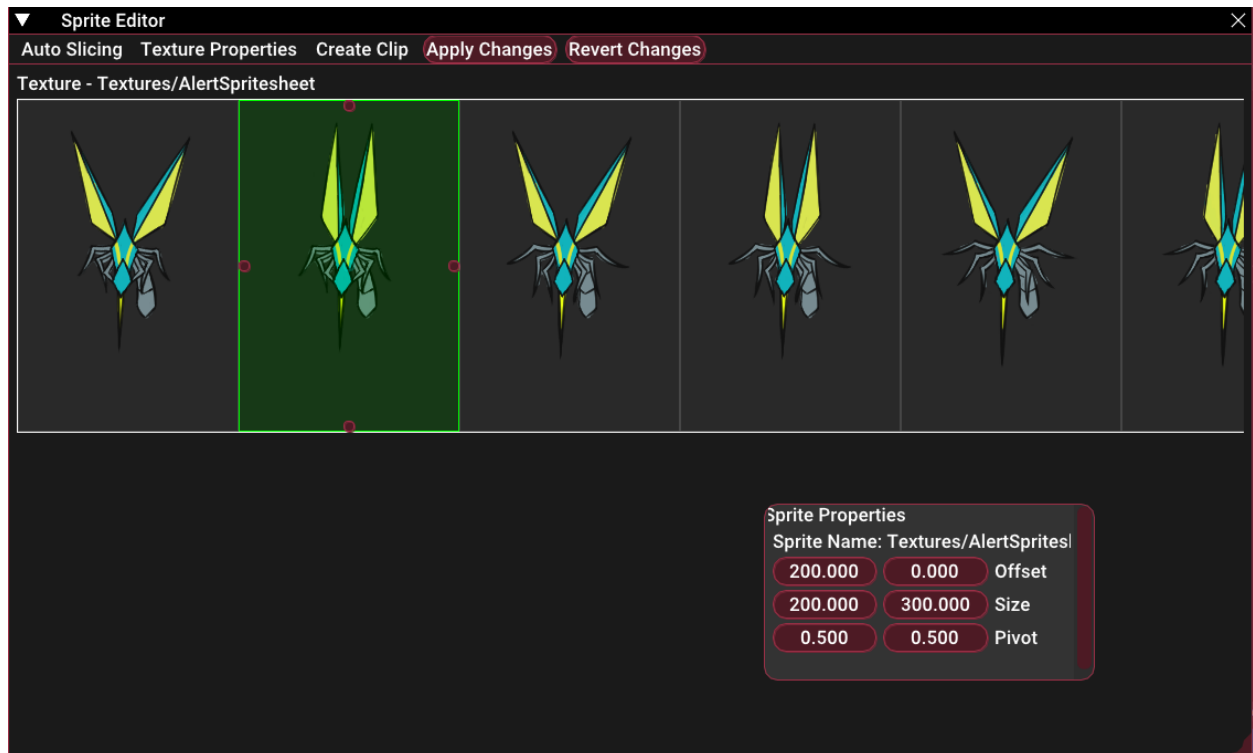
Sprite Editor Panel

The Sprite Editor is opened by double-clicking on a texture file inside the Asset Panel



- Drag & Drop
 - User can drag in another texture from the asset panel to edit that texture
- Auto Slicing
 - Automatic Slice Type: Will slice the sprite sheet based on the settings of pivot position and the toggled overlapping settings
 - Rows & Column Slice Type: Will slice the sprite sheet accordingly to the column and row with the indicated pivot position
- Texture Properties
 - Repeat Texture Wrap Mode: This texture will be repeated when used with tiling
 - Mirror Texture Wrap Mode: This texture will be mirrored when used with tiling
 - Clamp Texture Wrap Mode: This texture will be clamped and will be stretched when tiling
- Create Clip
 - Apply Changes and Create Clip: Create a clip with the indicated FPS & loop toggle. If the loop is checked, this clip will be looped.

Sprite Editor Panel



When you clicked on the individual sliced sprite you can do the following:

- Manual Slicing
 - Click on the points to modify the area of the slicing
- Sprite Properties
 - Modify the properties of the selected slicing
- Delete Selected Sprite
 - Pressing the key (Del) on the keyboard will delete the current selected slice
- Apply Changes
 - This will save the settings for this texture file
- Revert Changes
 - This will revert back to the original settings of the texture
- Highlight Slicing
 - Holding down (Ctrl) will highlight the slicing for this texture
- Zoom in / out
 - Holding down (Ctrl) and scrolling the mouse wheel will zoom in and out

Tilemap Panel

The tilemap panel is used to display all available tiles found in the Tilemap folder that can be used when drag into the world to create entities with the dragged texture.



Drag & Drop texture from file explorer

- Copy the texture into tilemap folder

Drag & Drop texture from asset panel

- Copy the texture into the tilemap folder

Drag & Drop from Tilemap Panel into the Viewport

- Create an entity with the selected texture

Viewport Panel

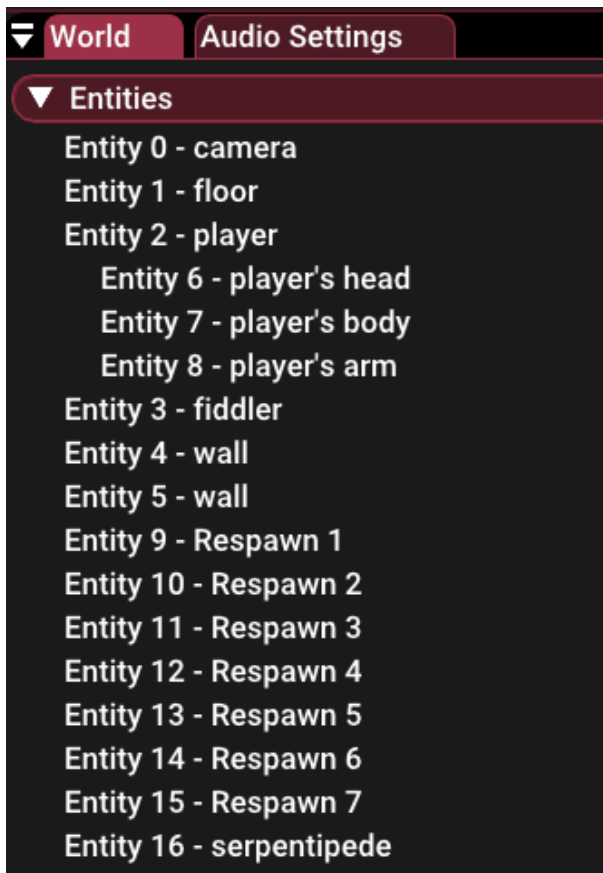


The viewport panel is the editor's camera, which is used to create the game world.

- Drag & Drop from Asset Panel / Tilemap Panel:
 - Create an entity with the dragged texture
- Moving the camera (Mouse cursor must be in the Viewport Panel to work)
 - Using keyboard, I/J/K/L
 -  Hand tool, hold the left mouse button and drag to move the camera
- Zooming in the camera (Mouse cursor must be in the Viewport Panel to work)
 - Using keyboard, U/O to zoom in and out
 - Using mouse scroll wheel
- Select Entity ( Selection tool must be selected)
 - Click to select the entity where the cursor is pointing at
 - Click to select the next entity behind the current selection

World Panel

The world panel displays the list of all entities that currently exist in the world, and the name that is assigned to each entity. Single clicking on an entity will open its property inspector, while double clicking on an entity will move the viewport camera to the entity's Transform position.



Parent Hierarchy

- Entities can be dragged into another entity
- Only supports 1 level of hierarchy

Debugging Information

Ctrl + Enter will print the debugging information into the console